

Multiply Two Numbers Without Using * Operator

Algorithm and Flowchart Correction & Verification

1. Mistakes in the Given Algorithm

- The algorithm always uses **num2** as the repetition count, so it does not guarantee the minimum possible number of additions.
- It does not compare $|\text{num1}|$ and $|\text{num2}|$ to choose the smaller absolute value as the repetition count.
- The condition **index** $\leq$ **num2** does not work correctly when num2 is negative.
- It does not correctly handle all combinations of positive and negative inputs.
- Zero is not handled explicitly, although multiplication by zero needs no additions.

2. Corrected Algorithm

1. Start.
2. Input num1 and num2.
3. If $\text{num1} = 0$ OR $\text{num2} = 0$, set $\text{prod} = 0$ and go to Print prod.
4. Find $|\text{num1}|$ and $|\text{num2}|$.
5. Set **addNum** = larger absolute value.
6. Set **count** = smaller absolute value.
7. Set $\text{prod} = 0$ and $\text{index} = 1$.
8. While $\text{index} \leq \text{count}$: $\text{prod} = \text{prod} + \text{addNum}$; $\text{index} = \text{index} + 1$.
9. If exactly one original input is negative, set $\text{prod} = -\text{prod}$.
10. Print prod.
11. Stop.

Minimum addition count = $\min(|\text{num1}|, |\text{num2}|)$

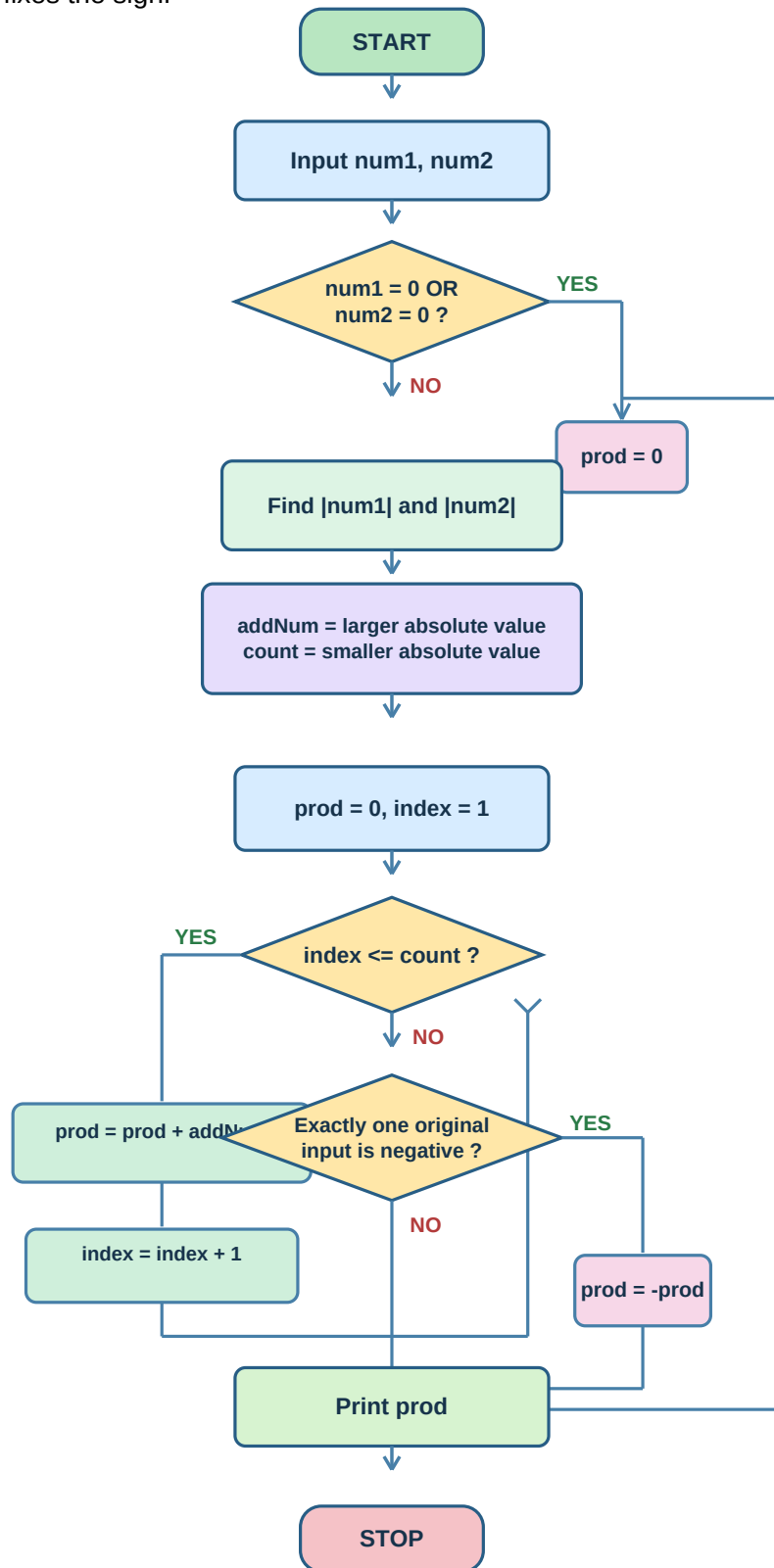
Example: $3 \times 12 = 12 + 12 + 12 = 36$, so only 3 additions are needed.

3. Mistakes in the Given Flowchart

- The flowchart directly uses num2 as the loop limit.
- It does not compare $|\text{num1}|$ and $|\text{num2}|$ to get the minimum addition count.
- It has no proper logic for negative inputs.
- It does not explicitly handle multiplication by zero.
- For a negative num2, $\text{index} \leq \text{num2}$ may immediately be false and give a wrong result.

4. Corrected Flowchart

Follow the arrows from top to bottom. The YES branch loops only for repeated addition; the NO branch exits the loop and fixes the sign.



5. Verification Using the Given Values

Test	Input (num1, num2)	Product	Minimum Additions
1	12, 3	36	3
2	3, 12	36	3
3	-6, 5	-30	5
4	6, -5	-30	5
5	-9, -4	36	4
6	-4, -9	36	4
7	0, 8	0	0
8	8, 0	0	0
9	0, -7	0	0
10	-7, 0	0	0
11	0, 0	0	0
12	1, -7	-7	1
13	-1, -9	9	1

6. Result

Result: The corrected algorithm and flowchart successfully multiply two numbers without using the * operator. They work for positive numbers, negative numbers, and zero while using the minimum possible number of additions. **Minimum Addition Count = $\min(|\text{num1}|, |\text{num2}|)$.**

Conclusion: The corrected method handles positive numbers, negative numbers, and zero while using the minimum number of repeated additions.